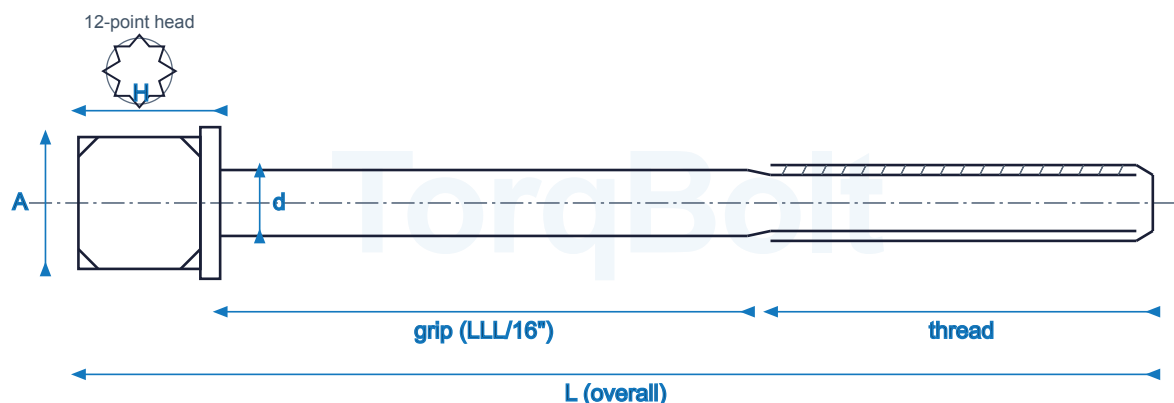


D I M E N S I O N A L D A T A S H E E T

MS21250

External Wrenching
12-Point
IMPERIAL

MS21250 / NASM21250 — 12-Point External Wrenching Tension Bolt · part number MS21250[H]-DD-LLL

TENSILE (FTU)	SHEAR (FSU)	HARDNESS	MAX TEMP	THREAD
180 ksi	108 ksi	39–43 HRC	450°F	UNJF-3A

Dia. dash (DD)	Nominal diameter	Thread series	H prefix	Commonly stocked grip dash codes (LLL — grip in 1/16")
03	0.190" (#10)	UNJF-3A	drilled = safety wire	04, 06, 08, 10, 12, 16
04	1/4"	1/4-28	H = drilled head	04, 06, 08, 10, 12, 13, 14, 16, 18, 20, 22, 24, 26, 28, 30, 34, 38, 40, 42, 46, 52, 54
05	5/16"	5/16-24	H = drilled head	06, 08, 09, 10, 12, 14, 15, 16, 17, 18, 22, 24, 26, 28, 30, 32, 36, 40, 48, 56, 62
06	3/8"	3/8-24	H = drilled head	06, 08, 10, 12, 14, 16, 18, 20, 22, 24, 26, 30, 34, 38, 40, 42, 44, 56, 62
07	7/16"	7/16-20	H = drilled head	04, 08, 14, 16, 20, 22, 30, 64
08	1/2"	1/2-20	H = drilled head	18, 20, 24, 26, 27, 28, 31, 32, 34, 38, 42, 47, 50, 62, 76
09	9/16"	9/16-18	H = drilled head	14, 18, 20, 22, 24, 38, 66
10	5/8"	5/8-18	H = drilled head	22, 24, 70
12	3/4"	3/4-16	H = drilled head	24, 36, 40
14	7/8"	7/8-14	H = drilled head	46

Grip dash → inches: divide by 16 (e.g. 022 = 1.375"); odd codes carry a +0.061" offset, so confirm odd-grip lengths against the controlling drawing. Dash 03 = 0.190" (#10), not 3/16". Material: aircraft-quality alloy steel (4140, 4340/E4340, 6150, 8735, 8740). Finish: cadmium plate + chromate per (AMS-)QQ-P-416 Type II Class 2. Standards: MS21250 = NASM21250 (rev L); procurement MIL-B-8831B (cancelled 2000) = NASM8831. Per-size head dimensions and shear/tensile loads in pounds are taken from the controlling NASM21250 drawing.